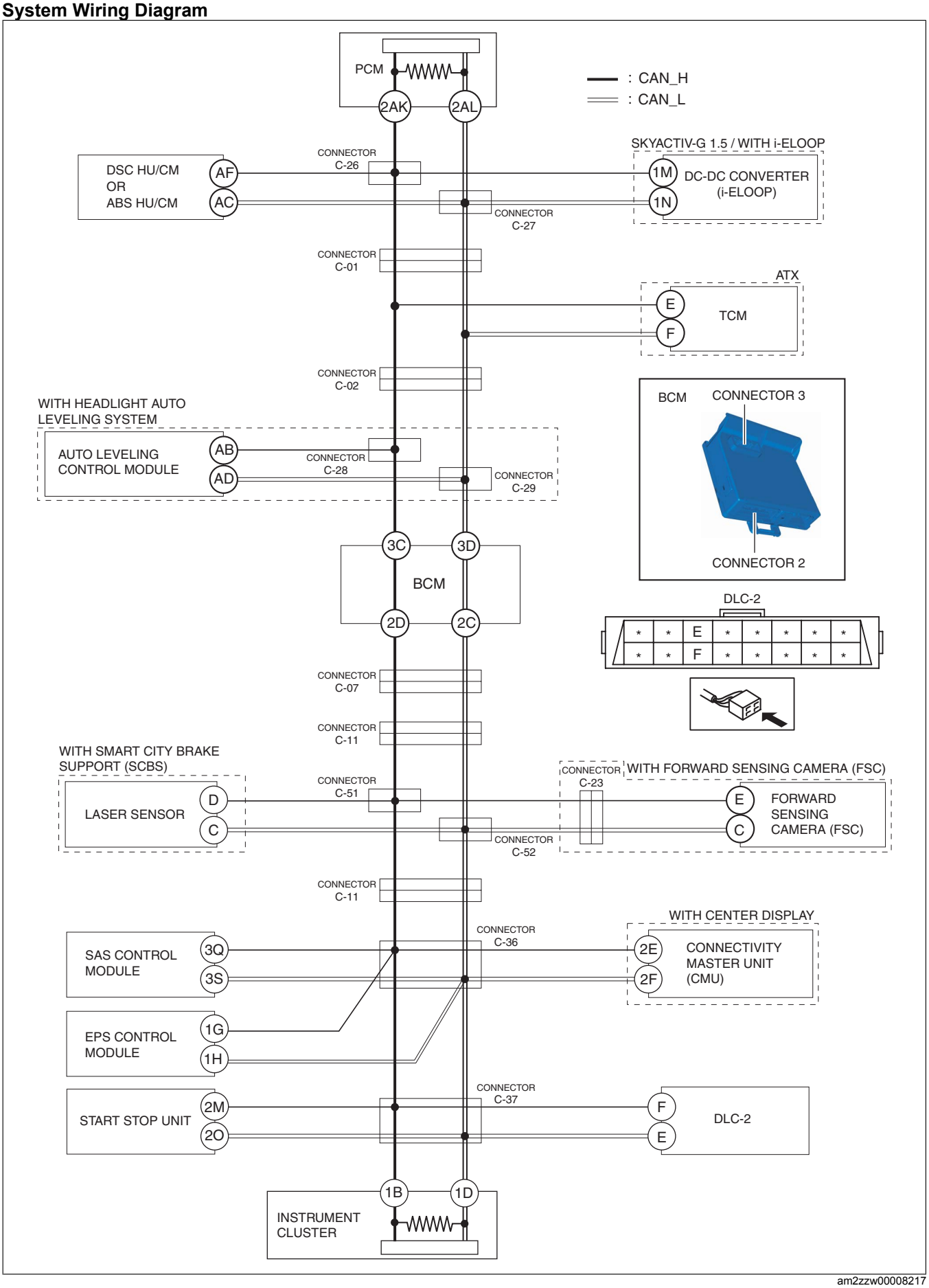
DETERMINING SHORT TO GROUND LOCATION (HS-CAN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)]

id100220000600

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a short to ground by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)**

System Wiring Diagram



am2zzw00008217

Determination Procedure

Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**
- **When inspecting the DLC-2, touch it with a paper clip or similar thin pin without directly inserting a tester into the terminals.**

Step	Inspection		Action
1	**INSPECT FOR SHORT TO GROUND BETWEEN BCM AND INSTRUMENT CLUSTER** • Switch the ignition off. • DISCONNECT the Negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION) • Disconnect the connector 2 and 3 which has BCM terminals 3C,3D and 2D,2C. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Go to Step 15.
		No	Go to the next step.
2	**INSPECT CAN LINE IN BCM FOR SHORT TO GROUND** • Inspect for continuity at the following terminals: — Between BCM terminal 3C and body ground — Between BCM terminal 3D and body ground • Is there continuity?	Yes	Replace the BCM because there is a short to ground in the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Go to the next step.
3	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-28 AND C-29 AND BCM** • Connect the connector 2 and 3 which has BCM terminals 3C,3D and 2D,2C. • Disconnect the connectors C-28 and C-29. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connectors C-28 and C-29 and BCM because the wiring harness is shorted to ground.
		No	Go to the next step.
4	**INSPECT FOR SHORT TO GROUND BETWEEN AUTO LEVELING CONTROL MODULE AND CONNECTORS C-28 AND C-29** • Inspect for continuity at the following terminals: — Between auto leveling control module terminal AB and body ground — Between auto leveling control module terminal AD and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 6.

Step	Inspection		Action
5	**INSPECT CAN LINE IN AUTO LEVELING CONTROL MODULE FOR SHORT TO GROUND** • Disconnect the auto leveling control module connector. • Inspect for continuity at the following terminals: — Between auto leveling control module terminal AB (wiring harness side) and body ground — Between auto leveling control module terminal AD (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the auto leveling control module and connectors C-28 and C-29 because the wiring harness is shorted to ground.
		No	Replace the auto leveling control module because there is a short to ground in the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/ INSTALLATION.)
6	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-02 AND CONNECTORS C-28 AND C-29** • Disconnect the connector C-02. • Connect the connectors C-28 and C-29. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connector C-02 and connectors C-28 and C-29 because the wiring harness is shorted to ground.
		No	Go to the next step.
7	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-01 AND CONNECTOR C-02** • Disconnect the connector C-01. • Connect the connector C-02. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 9.
8	**INSPECT CAN LINE IN TCM FOR SHORT TO GROUND** • Disconnect the TCM connector. • Inspect for continuity at the following terminals: — Between TCM terminal E (wiring harness side) and body ground — Between TCM terminal F (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the TCM and connector C-01,C-02 because the wiring harness is shorted to ground.
		No	Replace the TCM because there is a short to ground in the TCM. (See CONTROL VALVE BODY REMOVAL/ INSTALLATION [CW6A-EL].) (See CONTROL VALVE BODY REMOVAL/ INSTALLATION [EW6A-EL].)
9	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-26 AND C-27 AND CONNECTOR C-01** • Disconnect the connectors C-26 and C-27. • Connect the connector C-01. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connectors C-26 and C-27 and connector C-01 because the wiring harness is shorted to ground.
		No	Go to the next step.
10	**INSPECT FOR SHORT TO GROUND BETWEEN DSC HU/CM OR ABS HU/CM AND CONNECTORS C-26 AND C-27** • Inspect for continuity at the following terminals: — Between DSC HU/CM or ABS HU/CM terminal AF and body ground — Between DSC HU/CM or ABS HU/CM terminal AC and body ground • Is there continuity?	Yes	Go to the next step.
		No	• Go to Step 12.(SKYACTIV-G 1.5) • Go to Step 14.(SKYACTIV-G 1.3)

Step	Inspection		Action
11	**INSPECT CAN LINE IN DSC HU/CM OR ABS HU/CM FOR SHORT TO GROUND** • Disconnect the DSC HU/CM or ABS HU/CM connector. • Inspect for continuity at the following terminals: — Between DSC HU/CM or ABS HU/CM terminal AF (wiring harness side) and body ground — Between DSC HU/CM or ABS HU/CM terminal AC (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the DSC HU/CM or ABS HU/CM and connectors C-26 and C-27 because the wiring harness is shorted to ground.
		No	Replace the DSC HU/CM or ABS HU/CM because there is a short to ground in the DSC HU/CM or ABS HU/CM. (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].) (See ABS HU/CM REMOVAL/INSTALLATION [R.H.D.].)
12	**INSPECT FOR SHORT TO GROUND BETWEEN DC-DC CONVERTER (i-ELOOP) AND CONNECTORS C-26 AND C-27** • Inspect for continuity at the following terminals: — Between DC-DC converter (i-ELOOP) terminal 1M and body ground — Between DC-DC converter (i-ELOOP) terminal 1N and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 14.
13	**INSPECT CAN LINE IN DC-DC CONVERTER (i-ELOOP) FOR SHORT TO GROUND** • Disconnect the DC-DC converter (i-ELOOP) connector. • Inspect for continuity at the following terminals: — Between DC-DC converter (i-ELOOP) terminal 1M (wiring harness side) and body ground — Between DC-DC converter (i-ELOOP) terminal 1N (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the DC-DC converter (i-ELOOP) and connectors C-26 and C-27 because the wiring harness is shorted to ground.
		No	Replace the DC-DC converter (i-ELOOP) because there is a short to ground in the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
14	**INSPECT CAN LINE IN PCM FOR SHORT TO GROUND** • Disconnect the PCM connector. • Inspect for continuity at the following terminals: — Between PCM terminal 2AK (wiring harness side) and body ground — Between PCM terminal 2AL (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the PCM and connectors C-26 and C-27 because the wiring harness is shorted to ground.
		No	Replace the PCM because there is a short to ground in the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
15	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-07 AND INSTRUMENT CLUSTER** • Disconnect the connector C-07. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-07 and BCM because the wiring harness is shorted to ground.
16	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-07 AND CONNECTOR C-11** • Connect the connector C-07. • Disconnect the connector C-11. • Inspect for continuity at the following terminals: — Between BCM terminal 2D and body ground — Between BCM terminal 2C and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connector C-07 and connector C-11 because the wiring harness is shorted to ground.
		No	Go to the next step.

Step	Inspection		Action
17	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-11 AND INSTRUMENT CLUSTER** • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Go to Step 24.
		No	Go to the next step.
18	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-51 AND C-52 AND CONNECTOR C-11** • Connect the connector C-11. • Disconnect the connectors C-51 and C-52. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connectors C-51 and C-52 and connector C-11 because the wiring harness is shorted to ground.
		No	Go to the next step.
19	**INSPECT FOR SHORT TO GROUND BETWEEN LASER SENSOR AND CONNECTORS C-51 AND C-52** • Inspect for continuity at the following terminals: — Between laser sensor terminal D and body ground — Between laser sensor terminal C and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 21.
20	**INSPECT CAN LINE IN LASER SENSOR FOR SHORT TO GROUND** • Disconnect the laser sensor connector. • Inspect for continuity at the following terminals: — Between laser sensor terminal D (wiring harness side) and body ground — Between laser sensor terminal C (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the laser sensor and connectors C-51 and C-52 because the wiring harness is shorted to ground.
		No	Replace the laser sensor because there is a short to ground in the laser sensor. (See LASER SENSOR REMOVAL/INSTALLATION.)
21	**INSPECT FOR SHORT TO GROUND BETWEEN FORWARD SENSING CAMERA (FSC) AND CONNECTORS C-51 AND C-52** • Inspect for continuity at the following terminals: — Between forward sensing camera (FSC) terminal E and body ground — Between forward sensing camera (FSC) terminal C and body ground • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-11 and connectors C-51 and C-52 because the wiring harness is shorted to ground.
22	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-51 AND C-52 AND CONNECTOR C-23** • Disconnect the connector C-23. • Inspect for continuity at the following terminals: — Between forward sensing camera (FSC) terminal E and body ground — Between forward sensing camera (FSC) terminal C and body ground • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connectors C-51 and C-52 and connector C-23 because the wiring harness is shorted to ground.

Step	Inspection		Action
23	**INSPECT CAN LINE IN FORWARD SENSING CAMERA (FSC) FOR SHORT TO GROUND** • Disconnect the forward sensing camera (FSC) connector. • Inspect for continuity at the following terminals: — Between forward sensing camera (FSC) terminal E (wiring harness side) and body ground — Between forward sensing camera (FSC) terminal C (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the forward sensing camera (FSC) and connector C-23 because the wiring harness is shorted to ground.
		No	Replace the forward sensing camera (FSC) because there is a short to ground in the forward sensing camera (FSC). (See FORWARD SENSING CAMERA (FSC) REMOVAL/ INSTALLATION.)
24	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-36 AND INSTRUMENT CLUSTER** • Disconnect the connector C-36. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Go to Step 31.
		No	Go to the next step.
25	**INSPECT FOR SHORT TO GROUND BETWEEN SAS CONTROL MODULE AND CONNECTOR C-36** • Inspect for continuity at the following terminals: — Between SAS control module terminal 3Q and body ground — Between SAS control module terminal 3S and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 27.
26	**INSPECT CAN LINE IN SAS CONTROL MODULE FOR SHORT TO GROUND** • Disconnect the SAS control module connector. • Inspect for continuity at the following terminals: — Between SAS control module terminal 3Q (wiring harness side) and body ground — Between SAS control module terminal 3S (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the SAS control module and connector C-36 because the wiring harness is shorted to ground.
		No	Replace the SAS control module because there is a short to ground in the SAS control module. (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See SAS CONTROL MODULE REMOVAL/ INSTALLATION [STANDARD DEPLOYMENT CONTROL SYSTEM].)
27	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTIVITY MASTER UNIT (CMU) AND CONNECTOR C-36** • Inspect for continuity at the following terminals: — Between connectivity master unit (CMU) terminal 2E and body ground — Between connectivity master unit (CMU) terminal 2F and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 29.
28	**INSPECT CAN LINE IN CONNECTIVITY MASTER UNIT(CMU) FOR SHORT TO GROUND** • Disconnect the connectivity master unit (CMU) connector. • Inspect for continuity at the following terminals: — Between connectivity master unit (CMU) terminal 2E (wiring harness side) and body ground — Between connectivity master unit (CMU) terminal 2F (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connectivity master unit (CMU) and connector C-36 because the wiring harness is shorted to ground.
		No	Replace the connectivity master unit (CMU) because there is a short to ground in the connectivity master unit (CMU). (See CONNECTIVITY MASTER UNIT (CMU) REMOVAL/ INSTALLATION.)

Step	Inspection		Action
29	**INSPECT FOR SHORT TO GROUND BETWEEN EPS CONTROL MODULE AND CONNECTOR C-36** • Inspect for continuity at the following terminals: — Between EPS control module terminal 1G and body ground — Between EPS control module terminal 1H and body ground • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-11 and connector C-36 because the wiring harness is shorted to ground.
30	**INSPECT CAN LINE IN EPS CONTROL MODULE FOR SHORT TO GROUND** • Disconnect the EPS control module connector. • Inspect for continuity at the following terminals: — Between EPS control module terminal 1G (wiring harness side) and body ground — Between EPS control module terminal 1H (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the EPS control module and connector C-36 because the wiring harness is shorted to ground.
		No	Replace the EPS control module because there is a short to ground in the EPS control module. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
31	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-37 AND DLC-2** • Disconnect the connector C-37. • Inspect for continuity at the following terminals: — Between DLC-2 terminal F and body ground — Between DLC-2 terminal E and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the connector C-37 and DLC-2 because the wiring harness is shorted to ground.
		No	Go to the next step.
32	**INSPECT FOR SHORT TO GROUND BETWEEN START STOP UNIT AND CONNECTOR C-37** • Inspect for continuity at the following terminals: — Between start stop unit terminal 2M and body ground — Between start stop unit terminal 2O and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 34.
33	**INSPECT CAN LINE IN START STOP UNIT FOR SHORT TO GROUND** • Disconnect the start stop unit connector. • Inspect for continuity at the following terminals: — Between start stop unit terminal 2M (wiring harness side) and body ground — Between start stop unit terminal 2O (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the start stop unit and connector C-37 because the wiring harness is shorted to ground.
		No	Replace the start stop unit because there is a short to ground in the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.)
34	**INSPECT FOR SHORT TO GROUND BETWEEN INSTRUMENT CLUSTER AND CONNECTOR C-37** • Inspect for continuity at the following terminals: — Between instrument cluster terminal 1B and body ground — Between instrument cluster terminal 1D and body ground • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-36 and connector C-37 because the wiring harness is shorted to ground.
35	**INSPECT CAN LINE IN INSTRUMENT CLUSTER FOR SHORT TO GROUND** • Disconnect the instrument cluster connector. • Inspect for continuity at the following terminals: — Between instrument cluster terminal 1B (wiring harness side) and body ground — Between instrument cluster terminal 1D (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the instrument cluster and connector C-37 because the wiring harness is shorted to ground.
		No	Replace the instrument cluster because there is a short to ground in the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)